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ELECTRONIC DEVICE AND METHOD FOR IMAGE VIEWING

Abstract

An electronic device and a method for image viewing are provided. The method includes: obtaining a plurality of images, wherein the plurality of images include a first image, wherein the first image includes a first value and a second value; generating a directory of data by assigning the first image to a first directory of the directory of data and assigning the first image to a first subdirectory of the first directory according to the second value; in response to the first image being selected, preloading a second image within a second directory of the directory of data to a cache memory according to the directory of data; and outputting a graphical user interface, wherein the graphical user interface displays the first image in response to the first image being selected.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the priority benefit of U.S. provisional application Ser. No. 63/560,758, filed on Mar. 3, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure is related to image processing technology, and particularly related to an electronic device and a method for image viewing.

Description of Related Art

[0003] When using an image viewer, displaying multiple images simultaneously in one page may often result in significant loading times. As users switch between pages by the image viewer, the loading process may cause delays, leading to a poor user experience due to interruptions. As improving the speed and smoothness of image transitions is critical for enhancing overall user satisfaction, reducing image loading times has become a key challenge in the field.

SUMMARY

[0004] The disclosure is directed to an electronic device and a method for image viewing.

[0005] The present invention is directed to an electronic device for image viewing, including a transceiver, a cache memory, and a processor. The processor is coupled to the transceiver and the cache memory, wherein the processor is configured to: obtain a plurality of images through the transceiver, wherein the plurality of images include a first image, wherein the first image includes a first value and a second value; generate a directory of data by assigning the first image to a first directory of the directory of data according to the first value and assigning the first image to a first subdirectory of the first directory according to the second value; in response to the first image being selected, preload a second image within a second directory of the directory of data to the cache memory according to the directory of data; and output a graphical user interface through the transceiver, wherein the graphical user interface displays the first image in response to the first image being selected.

[0006] In one embodiment of the present invention, the processor is further configured to: determine an offset between an order of the first directory in the directory of data and an order of the second directory in the directory of data; determine whether the offset is less than or equal to a first range limit; and in response to the offset being less than or equal to the first range limit, preload the second image within the second directory.

[0007] In one embodiment of the present invention, the processor is further configured to: preload a third image within a third directory of the directory of data to the cache memory, wherein an order of the third directory in the directory of data is previous to an order of the second directory in the directory of data.

[0008] In one embodiment of the present invention, the processor is further configured to: in response to the first image being selected, preload a fourth image within the first subdirectory to the cache memory according to the directory of data.

[0009] In one embodiment of the present invention, the processor is further configured to:

determine an offset between an order of the first image in the first subdirectory and an order of the fourth image in the first subdirectory; determine whether the offset is less than or equal to a second range limit; and in response to the offset being less than or equal to the second range limit, preload the fourth image.

[0010] In one embodiment of the present invention, the processor is further configured to: output a graphical user interface through the transceiver, wherein the graphical user interface displays a plurality of videos according to a count value.

[0011] In one embodiment of the present invention, the processor is further configured to: determine the second range limit according to the count value.

[0012] In one embodiment of the present invention, the processor is further configured to: in response to the offset being less than the second range limit but the fourth image being the first or the last one among images within the first subdirectory or in response to the first image being the first or the last one among the images within the first subdirectory, save a preloading quota; determine a number of preloaded images of the first directory; and in response to the number of the preloaded images being less than or equal to a count limit, preload the second images within the second directory by the preloading quota.

[0013] In one embodiment of the present invention, the processor is further configured to: preload a fifth image within the first subdirectory to the cache memory after the fourth image being preloaded, wherein an order of the fifth image in the first subdirectory is previous to an order of the fourth image in the first subdirectory.

[0014] In one embodiment of the present invention, the first image further includes a third value, wherein the processor is further configured to: sort the first image in the first subdirectory according to the third value to generate the directory of data.

[0015] The present invention is directed to a method for image viewing, including: obtaining a plurality of images, wherein the plurality of images include a first image, wherein the first image includes a first value and a second value; generating a directory of data by assigning the first image to a first directory of the directory of data and assigning the first image to a first subdirectory of the first directory according to the second value; in response to the first image being selected, preloading a second image within a second directory of the directory of data to a cache memory according to the directory of data; and outputting a graphical user interface, wherein the graphical user interface displays the first image in response to the first image being selected.

[0016] In one embodiment of the present invention, the step of preloading the second image within the second directory of the directory of data to the cache memory according to the directory of data including: determining an offset between an order of the first directory in the directory of data and an order of the second directory in the directory of data; determine whether the offset is less than or equal to a first range limit; and in response to the offset being less than or equal to the first range limit, preloading the second image within the second directory.

[0017] In one embodiment of the present invention, the method further including: preloading a third image within a third directory of the directory of data to the cache memory, wherein an order of the third directory in the directory of data is previous to an order of the second directory in the directory of data.

[0018] In one embodiment of the present invention, the method further including: in response to the first image being selected, preloading a fourth image within the first subdirectory to the cache memory according to the directory of data.

[0019] In one embodiment of the present invention, the step of preloading the fourth image within the first subdirectory to the cache memory according to the directory of data including: determining an offset between an order of the first image in the first subdirectory and an order of the fourth image in the first subdirectory; and determining whether the offset is less than or equal to a second range limit; and in response to the offset being less than or equal to the second range limit, preloading the fourth image.

[0020] In one embodiment of the present invention, the method further including: outputting a graphical user interface, wherein the graphical user interface displays a plurality of videos according to a count value.

[0021] In one embodiment of the present invention, the method further including: determining the second range limit according to the count value.

[0022] In one embodiment of the present invention, the step of preloading the second image within the second directory of the directory of data to the cache memory according to the directory of data including: in response to the offset being less than the second range limit but the fourth image being the first or the last one within the first subdirectory or in response to the first image being the first or the last one among the images within the first subdirectory, saving a preloading quota; determining a number of preloaded images of the first directory; and in response to the number of the preloaded images being less than or equal to a count limit, preloading the second images within the second directory by the preloading quota.

[0023] In one embodiment of the present invention, the method further including: preloading a fifth image within the first subdirectory to the cache memory after the fourth image being preloaded, wherein an order of the fifth image in the first subdirectory is previous to an order of the fourth image in the first subdirectory.

[0024] In one embodiment of the present invention, the first image further includes a third value, wherein the method further includes: sorting the first image in the first subdirectory according to the third value to generate the directory of data.

[0025] Based on the above description, the disclosure designs a special format for image to create a directory of data. The electronic device may preload a plurality of images based on image currently being viewed by the user.

[0026] To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0028] FIG. 1 illustrates a schematic diagram of an electronic device for image viewing according to one embodiment of the present disclosure.

[0029] FIG. 2 illustrates a schematic diagram of a format or an image file according to one embodiment of the present disclosure.

[0030] FIG. 3 illustrates data structure of image files according to one embodiment of the present disclosure.

[0031] FIG. 4 illustrates a graphical user interface of an image viewer according to one embodiment of the present disclosure.

[0032] FIG. 5 illustrates a flowchart for image preloading according to one embodiment of the present disclosure.

[0033] FIG. 6 illustrates an example of image preloading according to one embodiment of the present disclosure.

[0034] FIG. 7 illustrates another example of image preloading according to one embodiment of the present disclosure.

[0035] FIG. 8 illustrates another example of image preloading according to one embodiment of the present disclosure.

[0036] FIG. **9** illustrates a flowchart of a method for image viewing according to one embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0037] FIG. **1** illustrates a schematic diagram of an electronic device **100** for image viewing according to one embodiment of the present disclosure. The electronic device **100** may include a processor **110**, a storage medium **120**, and a transceiver **130**. The processor **110** may be, for example, a central processing unit (CPU), or other programmable general purpose or special purpose micro control unit (MCU), a microprocessor, a digital signal processor (DSP), a programmable controller, an application specific integrated circuit (ASIC), a graphics unit (GPU), an arithmetic logic unit (ALU), a complex programmable logic device (CPLD), a field programmable gate array (FPGA), or other similar device or a combination of the above devices. The processor **110** may be coupled to the storage medium **120** and the transceiver **130**.

[0038] The storage medium **120** may be, for example, any type of fixed or removable random-access memory (RAM), a read-only memory (ROM), a flash memory, a hard disk drive (HDD), a solid-state drive (SSD) or similar element, or a combination thereof. The storage medium **120** may be a non-transitory computer readable storage medium configured to record a plurality of executable computer programs, modules, or applications to be loaded by the processor **110** to perform the functions of the electronic device **100**. The storage medium **120** may include a cache memory **121**. The processor **110** may preload one or more videos (or image sets) or images to the cache memory for image viewing purpose.

[0039] The transceiver **130** may be configured to transmit or receive wired/wireless signals. The transceiver **130** may also perform operations such as low noise amplifying, impedance matching, frequency mixing, up or down frequency conversion, filtering amplifying, and so forth. The processor **110** may communicate with other devices via the transceiver **130**. For example, the processor **110** may obtain one or more videos, images, or user commands through the transceiver **130**. For another example, the processor **110** may output a graphical user interface (GUI) to a display through the transceiver **130**.

[0040] FIG. **2** illustrates a schematic diagram of a format or an image file according to one embodiment of the present disclosure. The image **20** may include information **21** which may be defined as level 1 information, information **22** which may be defined as level 2 information, and information **23** which may be defined as level 3 information. The information **21** may include values such as a level_1 identification (ID) or a level_1 order. The information **22** may include values such as a level_2 ID or a level_2 order. The information **23** may include values such as a level_3 ID or a level_3 order.

[0041] One or more level_2 IDs may belong to the same level_1 ID, and one or more level_3 IDs may belong to the same level_2 ID. Take FIG. **3** as an example, the images **311** and **312** with different level_3 IDs but the same level_2 ID may belong to the same level_2 image group **310**. The image groups **310** and **320** with different level_2 IDs but the same level_3 ID may belong to the same level_1 image group **300**.

[0042] In one embodiment, the level_1 ID may represent an index of a directory in an operating system, wherein the corresponding level_1 order may represent the order of the level_1 ID among a plurality of level_1 IDs. The level_2 ID may represent an index of a subdirectory in the directory, wherein the corresponding level_2 order may represent the order of the level_2 ID among a plurality of level_2 IDs belonging to the same level_1 ID. The level_3 ID may represent an index of an image in the subdirectory, wherein the corresponding level_3 order may represent the order of the level_3 ID among a plurality of level_3 IDs belonging to the same level_2 ID.

[0043] For example, assume that the plurality of images obtained by the processor **110** are related to transthoracic echocardiography (TTE) images of patients. The images with the same level_1 ID may correspond to the same patient. The images with the same level_1 ID but different level_2 IDs may correspond to different electrodes attached on the same patient. That is, the images with

different level_2 IDs may correspond to different TTE videos of the same patient. The images with the same level_2 ID but different level_3 IDs may correspond to different frames (or time stamps, TTE images) of the same TTE video.

[0044] FIG. 4 illustrates a graphical user interface **400** of an image viewer according to one embodiment of the present disclosure. The GUI **400** may select and display one or more videos (or images) **410**, **420**, **430**, **440**, **450**, and **460** corresponding to the same level_1 ID simultaneously based on a count value (e.g., count value C) and the level_2 order of each video. For example, the GUI **400** may display 6 videos simultaneously since the count value is equal to 6, wherein the videos **410**, **420**, **430**, **440**, **450**, and **460** may be positioned on the GUI **400** according to level_2 order of each video. For example, the image **410** with the smallest level_2 order may be positioned in the upper-left corner of the GUI **400**, and the image **460** with the largest level_2 order may be positioned in the lower-right corner of the GUI **400**. In one embodiment, the displayed videos **410**, **420**, **430**, **440**, **450**, and **460** may correspond to different level_2 IDs but the same level_3 ID. That is, the time stamps of the different videos **410**, **420**, **430**, **440**, **450**, and **460** displayed simultaneously may be the same.

[0045] In one embodiment, the user may input a command to the electronic device **100** to manipulate the slider **42** of the GUI **400**, so as to change the time stamp of the displayed videos. For example, assume that the videos **410-460** displayed by the GUI **400** currently are corresponded to frame Y. If the user drags the slider **42** to the right, the GUI **400** may switch the displayed videos **410-460** from frame Y to frame Y+1.

[0046] In one embodiment, the user may input a command to the electronic device **100** to manipulate the bottom **41** of the GUI **400**, so as to change the videos being displayed. For example, assume that the videos **410-460**, corresponding to TTE acoustic windows **1-6** respectively, are being displayed by the GUI **400**. If the user clicks the bottom **41**, the GUI **400** may switch the displayed videos from videos **410-460** to videos corresponding to TTE acoustic windows **7-12**.

[0047] In one embodiment, if an image is selected (e.g., the image is displayed by the GUI **400** currently), the processor **110** may preload one or more images related to the selected image based on the parameters shown in Table 1.

TABLE-US-00001 TABLE 1 Level_1 Preload Range Limit: B (B is a positive integer and $0 \leq B \leq$ the maximum value of level_1 order. Level_3 Preload Range Limit (when $C = 1$): R (R is a positive integer and $0 \leq R \leq$ the maximum value of level_3 order. Level_3 Preload Range Limit (when $C > 1$): $L = \lfloor (0 - C)/2C \rfloor$ One Level_1 Group Preload Count limit: $O = 2R + 1$ All Level_1 Group Preload count limit: $A = 2O + 1 = 4R + 3$ Level_2 Count Value: C (C is a positive integer and $1 \leq C \leq$ the maximum value of level_2 order.

[0048] FIG. 5 illustrates a flowchart for image preloading according to one embodiment of the present disclosure, wherein the flowchart may be implemented by the electronic device **100** as shown in FIG. 1.

[0049] In step S501, the processor **110** may obtain a plurality of images through the transceiver **130**, wherein each image may include information such as information **21**, **22**, or **23** (i.e., level 1 information, level 2 information, or level 3 information) as shown in FIG. 2. The obtained images may be stored in the storage medium **120**.

[0050] In step S502, the processor **110** may generate a directory of data, wherein the directory of data may include one or more directory, and each directory may include one or more subdirectory. The processor **110** may generate the directory of data by assigning an image to a directory according to the information **21** and assigning the image to a subdirectory of the directory according to information **22**. The processor **110** may sort the image in the subdirectory according to information **23**.

[0051] For example, an image X may include values i, j, and k corresponding to information **21**, **22**, and **23**, respectively. The processor **110** may allocate the path X(i, j, k) of image X to the k-th image in the j-th subdirectory in the i-th directory of the directory of data. That is, index i may be

associated with level_1 ID or level_1 order of image X, index j may be associated with level_2 ID or level_2 order of image X, and index k may be associated with level_3 ID or level_3 order of image X. In one example, the path X(i, j, k) mentioned above may denote the k-th frame (or k-th time stamp) of the j-th TTE video (or j-th electrode) of the i-th patient.

[0052] In step S503, the user may select an image, wherein the selected image may be preloaded to the cache 121 or be displayed in the GUI 400. In the following paragraphs, It is assumed that the selected image is image X corresponding to path X(i, j, k).

[0053] In step S504, the processor 110 may preload one or more obtained images to the cache memory 121 according to the directory of data.

[0054] In one embodiment, the processor 110 may preload one or more images within the j-th subdirectory (e.g., the j-th TTE video of a patient) to the cache memory 121 according to the range limit R or L, wherein the processor 110 may determine which range limit, R or L, should be referred to according to the count value C. It should be noted that, in one embodiment, the number of preloaded images (including the image X(i, j, k) selected currently) belonging to the same level_1 ID cannot be greater than count limit $O=2R+1$, and the number of total preloaded images (including the image X(i, j, k)) cannot be greater than the count limit A, wherein the count limit A may be set to $A=2O+1=4R+3$ according to user defined.

[0055] Assume that $C=1$ (i.e., GUI 400 displays only one video or image in the same time). The processor 110 may determine an offset K between the level_3 order of the image X(i, j, k) in the j-th subdirectory and the level_3 order of an image X(i, j, $k\pm K$) in the j-th subdirectory, wherein K is a positive integer. If the offset $K\leq$ the range limit R, the processor 110 may preload the image X(i, j, $k\pm K$) to the cache 121. The order of the image X(i, j, $k\pm K$) in the j-th subdirectory to be preloaded may satisfy the following formula: $\max(1, k-R)\leq k\pm K\leq \min(k+R, \text{the largest level_3 order belonging to the same level_2 ID})$.

[0056] Take FIG. 6 as an example, assume that the count value $C=1$, the range limit $R=2$, and image 624 is currently selected or displayed by the GUI 400. The processor 110 may determine that the offsets K between the level_3 order (e.g., 4) of the image 624 and the level_3 orders (e.g., 2, 3, 5, and 6) of the images 622, 623, 625, and 626 are 2, 1, 1, and 2 respectively, which are less than or equal to the range limit $R=2$. Accordingly, the processor 110 may preload the images 622, 623, 625, or 626 to the cache 121. On the other hand, the processor 110 may determine that the offsets K between the level_3 order (e.g., 4) of the image 624 and the level_3 orders (e.g., 1 and 7) of the images 621 and 627 are 3 and 3 respectively, which are greater than the range limit $R=2$. Accordingly, the processor 110 may not preload the images 621 or 627 to the cache 121. The level_3 order $4\pm K$ of images to be preloaded (e.g., 622, 623, 625, or 626) may satisfy the following formula: $\max(1, 4-2)\leq 4\pm K\leq \min(4+2, 7)$.

[0057] Assume that $C>1$ (i.e., GUI 400 displays multiple videos or images in the same time). The processor 110 may determine an offset K between the level_3 order of the image X(i, j, k) in the j-th subdirectory and the level_3 order of an image X(i, j, $k+K$) in the j-th subdirectory, wherein K is a positive integer. If the offset $K\leq$ the range limit $L=[(O-C)/2C]$, the processor 110 may preload the image X(i, j, $k\pm K$) to the cache 121. The level_3 order of the image X(i, j, $k\pm K$) in the j-th subdirectory to be preloaded may satisfy the following formula: $\max(1, k-L)\leq k\pm K\leq \min(k+L, \text{the largest level_3 order belonging to the same level_2 ID})$.

[0058] Take FIG. 8 as an example, assume that the count value $C=2$, the range limit $R=3$, the count limit $O=2R+1=7$, the range limit $L=[(O-C)/2C]=1$, and the image 822 is currently selected or displayed by the GUI 400. The processor 110 may determine the offsets K between the level_3 order (e.g., 2) of the image 822 and the level_3 orders (e.g., 1 and 3) of the images 821 and 823 are 1 and 1, which are equal to the range limit $L=1$. Accordingly, the processor 110 may preload the images 821 and 823 to the cache 121. On the other hand, the processor 110 may determine that the offset K between the level_3 order (e.g., 2) of the image 822 and the level_3 order (e.g., 4) of the image 824 is 2, which is greater than the range limit $L=1$. Accordingly, the processor 110 may not

preload the image **824** to the cache **121**. The level_3 order $k+K$ of the images to be preloaded (e.g., **821** or **823**) may satisfy the following formula: $\max(1, 2-1) \leq 4+K \leq \min(2+1, 7)$.

[0059] In one embodiment, the processor **110** may preload the image $X(i, j, k-K)$ after the image $X(i, j, k+K)$ being preloaded in response to the level_3 order of the image $X(i, j, k-K)$ in the j -th subdirectory being previous to the level_3 order of the image $X(i, j, k+K)$ in the j -th subdirectory. Take FIG. 6 as an example, since the level_3 order of the image **623** is previous to the level_3 order of the image **625**, the processor **110** may preload the image **623** after the image **625** being preloaded.

[0060] In one embodiment, the processor **110** may preload the image $X(i, j, k+(K+n))$ after the image $X(i, j, k+K)$ being preloaded in response to the offset K between the level_3 order of the image $X(i, j, k+K)$ and the level_3 order of the image $X(i, j, k)$ is less than the offset $(K+n)$ between the level_3 order of the image $X(i, j, k+(K+n))$ and the level_3 order of the image $X(i, j, k)$, where n is a positive integer. Take FIG. 6 as an example, since the offset '1' between the level_3 order of the image **624** and the level_3 order of the image **623** is less than the offset '2' between the image **622** and the image **624**, the processor **110** may preload the image **622** after the image **623** being preloaded.

[0061] In one embodiment, if the offset K between the level_3 order of the image $X(i, j, k)$ and the level_3 order of the image $X(i, j, k \pm K)$ is less than the range limit (e.g., range limit R or L depending on the value of count value C), and the image $X(i, j, k \pm K)$ is not the first or the last one among images within the j -th subdirectory (i.e., the image $X(i, j, k \pm K)$ is not the image $X(i, j, 1)$ or $X(i, j, \text{the largest level_3 order (e.g., 7)})$), the processor **110** may preload the image $X(i, j, k \pm K)$. However, if the offset K between the level_3 order of the image $X(i, j, k)$ and the level_3 order of the image $X(i, j, k \pm K)$ is less than the range limit (e.g., range limit R or L), but the image $X(i, j, k \pm K)$ is the first or the last one among images within the j -th subdirectory, the processor **110** may save one or more preloading quotas for preloading the other image, wherein the number of the saved preloading quotas may be equal to the difference between the range limit R and the offset K (i.e., the number of the saved preloading quotas = $R - \text{offset } K$). After determining the images corresponding to the i -th directory to be preloaded, the processor **110** may determine the number of preloaded images of the i -th directory, wherein the more preloading quotas, the fewer the preloaded images. If the number of the preloaded images of the i -th directory is greater than or equal to the count limit $O = 2R + 1$, the processor **110** may determine that there no preloading quotas have been saved. If the number of the preloaded images of the i -th directory is less than the count limit O , the processor **110** may determine that one or more preloading quotas have been saved. The processor **110** may preload the image of the directory within the range limit B by the saved preloading quotas until all the preloading quotas are used.

[0062] Take FIG. 7 as an example, assume that the count value $C = 1$, the range limit $R = 3$, the range limit $B = 2$, the count limit $O = 7$, and image **722** is currently selected or displayed by the GUI **400**. The processor **110** may determine that the offset '1' between the level_3 order (e.g., 2) of the image **722** and the level_3 order (e.g., 1) of the image **721** is less than the range limit $R = 3$, but the image **721** is the first image within a subdirectory (i.e., the subdirectory includes images corresponding to the same level_2 ID or order) of the directory **720**. Accordingly, the processor **110** may save two preloading quotas for preloading the other image based on the formula: $R - \text{offset '1'} = 3 - 1 = 2$. The saved preloading quotas may be used for preloading the images within, for example, the directory **740** since the directory **740** is within the range limit B .

[0063] In one embodiment, the processor **110** may preload one or more images within the $(i \pm I)$ -th directory to the cache memory **121** according to the range limit B , wherein I is a positive integer and $I \leq \text{the range limit } B$. Specifically, the processor **110** may determine an offset I between the level_1 order of the i -th directory and the level_1 order of the $(i \pm I)$ -th directory. If the offset $I \leq B$, the processor **110** may preload the image $X(i+I, j, 1+K)$ in the $(i+I)$ -th directory and/or the image $X(i-I, j, m-K)$ in the $(i-I)$ -th directory, wherein m is the maximum level_3 order belonging to the

(i-I)-th directory, K is a non-negative integer, and $K \leq$ the range limit R. The level_1 order of the image X(i+I, j, 1+K) may satisfy the following formula: $i+I \leq \min(i+B$, the largest level_1 order belonging to the directory of data), and the level_3 order of the image X(i+I, j, 1+K) may satisfy the following formula: $1+K \leq \min(1+R$, the largest level_3 order belonging to the (i+I)-th directory). The level_1 order of the image X(i-I, j, m-K) may satisfy the following formula: $\max(1, i-B) \leq i-I$, and the level_3 order of the image X(i-I, j, m-K) may satisfy the following formula: $\max(1, m-R) \leq m-K$.

[0064] Take FIG. 6 as an example, assume that the range limit B=1 and image 624 is currently selected or displayed by the GUI 400. The processor 110 may determine the offset I between the level_1 order (e.g., 2) of the image 624 and the level_1 order (e.g., 3) of the image 631 (or 632, 633) is 1, which is less than or equal to the range limit B=1. Accordingly, the processor 110 may preload the image 631 (or 632, 633) to the cache 121. On the other hand, the processor 110 may determine the offset I between the level_1 order (e.g., 2) of the image 624 and the level_1 order (e.g., 1) of the images 617 (or 616, 615) is 1, which is less than or equal to the range limit B=1. Accordingly, the processor 110 may preload the image 617 (or 616, 615) to the cache 121.

[0065] In one embodiment, the processor 110 may preload the image X(i-I, j, m-K) after the image X(i+I, j, 1+K) being preloaded in response to the level_1 order of the image X(i-I, j, m-K) being previous to the level_3 order of the image X(i+I, j, 1+K), wherein m is the maximum level_3 order belonging to the (i-I)-th directory, K is a positive integer, and $K \leq$ the range limit R. Take FIG. 6 as an example, since the level_1 order of the images 617, 616, or 615 are previous to the level_1 order of the images 631, 632, or 633, the processor 110 may preload the images 617, 616, or 615 after the images 631, 632, or 633 being preloaded.

[0066] In one embodiment, assume that n and m are positive integer and $n < m$. The processor 110 may preload the image belonging to the (i+m)-th directory after the image belonging to the (i+n)-th directory being preloaded in response to the offset 'n' between the level 1 order of the (i+n)-th directory and level_1 order of the i-th directory is less than the offset 'm' between the level_1 order of the (i+m)-th directory and level_1 order of the i-th directory. Take FIG. 7 as an example, since the offset '1' between the directory 730 and the directory 720 is less than the offset '2' between the directory 740 and the directory 720, the processor 110 may preload the images within the directory 740 after the images within the directory 730 being preloaded.

[0067] In one embodiment, the processor 110 may determine the number of preloaded images. If the number of the preloaded images is less than the count limit $A=2O+1=4R+3$, the processor 110 may preload more images within the (i+n) directory, wherein n is a positive integer and $n \leq$ the range limit B. Take FIG. 7 as an example, assume that the range limit R=3, the range limit B=2, and the count limit A=15. After the images within the directory 710 being preloaded, the processor 110 may determine that the number of preloaded images (i.e., images 714, 715, 716, 717, 721, 722, 723, 724, 725, 731, 732, 733, and 734) within the directories 710, 720 and 730 is 13, which is less than the count limit A=15. Accordingly, the processor 110 may preload the images within the directory 740.

[0068] In one embodiment, the processor 110 may stop preloading the images if the difference between the count limit A and the number of the preloaded images is less than the count value C. Take FIG. 8 as an example. Assume that the count limit A=15 and the count value C=2, and the images (e.g., images 816 and 817 within one subdirectory of the directory 810, images 821, 822, and 823 within one subdirectory of the directory 820, or images 831 and 832 within one subdirectory of the directory 830) within two subdirectories of each directory 810, 820, and 830 are already preloaded to the cache 121. The processor 110 may determine that the number of the preloaded images is equal to '14'. Since the difference between the count limit A=15 and the number of the preloaded images '14' is less than the count value C=2, the processor 110 may stop preloading the images.

[0069] Back to FIG. 2, in step S505, the processor 110 may display one or more images through the

GUI **400**. The processor **110** may display one or more videos (or images) corresponding to the same level_1 ID based on the count value C and the level_2 order of each video.

[0070] In step **S506**, the processor **110** may determine whether the user switch the page of the GUI **400** through the user command, wherein each page may include one or more selected video (or images). If the page is switched, the processor **110** may select one or more new videos (or images) to be displayed on the GUI **400**. After the new videos (or images) are selected, the processor **110** may execute step **S503** again. If the page is not switched, the image preloading procedure may be stopped.

[0071] FIG. **6** illustrates an example of image preloading according to one embodiment of the present disclosure. Assume that the range limit $R=2$, the range limit $B=2$, the count value $C=1$, the count limit $O=2R+1=5$, the count limit $A=4R+3=11$, and the image **624** within the directory **620** is currently selected or displayed by the GUI **400**. The processor **110** may determine that the offsets between the level_3 order (e.g., 4) of the image **624** and the level_3 orders (e.g., 2, 3, 5, and 6) of the images **622**, **623**, **625**, and **626** are 2, 1, 1, and 2 respectively, which are less than or equal to the range limit $R=2$. Accordingly, the processor **110** may preload the images **622**, **623**, **625**, or **626** to the cache **121**. The processor **110** may preload the images **623** and **625** first since the offsets between the image **624** and the images **623** and **625** are the smallest offsets. The processor **110** may preload the images **625** first then preload the image **623** in response to the level_3 order of the images **623** being previous to the level_3 order of the image **625**. On the other hand, the processor **110** may determine that the offsets K between the level_3 order (e.g., 4) of the image **624** and the level_3 orders (e.g., 1 and 7) of the images **621** and **627** are 3 and 3 respectively, which are greater than the range limit $R=2$. Accordingly, the processor **110** may not preload the images **621** or **627** to the cache **121**.

[0072] The processor **110** may determine that the number of the preloaded images is '5', which is less than count limit $A=11$. Therefore, the processor **110** may preload more images within the directory other than the directory **620**. Since the level_1 order of the directory **610** is previous to the level_1 order of the directory **630**, the processor **110** may preload the images within the directory **630** first. The processor **110** may preload the images **631**, **632**, and **633** since the level_3 orders of those images satisfy the formula $1+K \leq \min(1+R, \text{the largest level}_3 \text{ order belonging to the } (i+1)\text{-th directory})$ as mentioned above.

[0073] The processor **110** may determine that the number of the preloaded images belonging to the directories **620** and **630** is '8', which is less than the count limit $A=11$. Therefore, the processor **110** may preload more images within the directory **610**. The processor **110** may preload the images **617**, **616**, and **615** since the level_3 orders of those images satisfy the formula $\max(1, m-R) \leq m-K$ as mentioned above.

[0074] FIG. **7** illustrates another example of image preloading according to one embodiment of the present disclosure. Assumed that the range limit $R=3$, the range limit $B=2$, the count value $C=1$, the count limit $O=2R+1=7$, the count limit $A=4R+3=15$, and the image **722** within the directory **720** is currently selected or displayed by the GUI **400**. The processor **110** may determine that the offsets between the level_3 order (e.g., 2) of the image **722** and the level_3 orders (e.g., 1, 3, 4, and 5) of the images **721**, **723**, **724**, and **725** are 1, 1, 2, and 3 respectively, which are less than or equal to the range limit $R=3$. Accordingly, the processor **110** may preload the images **721**, **723**, **724**, or **725** to the cache **121**. The processor **110** may preload the images **721** and **723** first since the offsets between the image **722** and the images **721** and **723** are the smallest offsets. The processor **110** may preload the image **723** first then preload the image **721** in response to the level_3 order of the image **721** being previous to the level_3 order of the image **723**.

[0075] The processor **110** may determine that the offset '1' between the level_3 order (e.g., 2) of the image **722** and the level_3 order (e.g., 1) of the image **721** is less than the range limit $R=3$, but the image **721** is the first image within a subdirectory (i.e., the subdirectory includes images corresponding to the same level_2 ID or order) of the directory **720**. Accordingly, the processor **110**

may save two preloading quotas for preloading the other image based on the formula: $R - \text{offset}$ '1' = $3 - 1 = 2$ as mentioned above.

[0076] The processor **110** may determine that the number of the preloaded images is '5', which is less than the count limit $A=15$. Therefore, the processor **110** may proceed to preload the other images. Since the offsets between the directory **720** and the directories **710** and **730** are the smallest offsets and within the range limit $B=2$, the processor **110** may select one of the directories **710** and **730** to preload the images. The processor **110** may preload the images within the directory **730** first in response to the level_1 order of the directory **710** being previous to the level_1 order of the directory **730**. The processor **110** may preload the images **731**, **732**, **733**, and **734** since the level_3 orders of those images satisfy the formula $1+K \leq \min(1+R, \text{the largest level}_3 \text{ order belonging to the } (i+I)\text{-th directory})$ as mentioned above. Then, the processor **110** may preload the images within the directory **710** since the number of the preloaded images is '9', which is less than the count value $A=15$. The processor **110** may preload the images **717**, **716**, **715**, and **714** since the level_3 orders of those images satisfy the formula $\max(1, m-R) \leq m-K$ as mentioned above.

[0077] The processor **110** may determine the number of the preloaded images within the directories **710**, **720**, and **730** is '13', which is less than the count limit $A=15$. Therefore, the processor **110** may preload the images within the range limit B by using the saved preloading quotas. For example, the processor **110** may preload two images within the directory **740** by using two preloading quotas.

[0078] FIG. 8 illustrates another example of image preloading according to one embodiment of the present disclosure. Assume that the range limit $R=3$, the range limit $B=1$, the count value $C=2$, the count limit $O=2R+1=7$, the count limit $A=4R+3=15$, and the two images (e.g., image **822** and another image shares the same level_3 and level_1 order but has a different level_2 order) respectively corresponding to the two subdirectories of the directory **820** are currently selected or displayed by the GUI **400**. Since the count value $C>1$, the processor **110** may determine the range limit $L=[(O-C)/2C]=1$. The processor **110** may determine the offsets between the level_3 order (e.g., 2) of the image **822** and the level_3 orders (e.g., 1 and 3) of the images **821** and **823** are 1 and 1, which are equal to the range limit $L=1$. Accordingly, the processor **110** may preload the images **821** and **823** to the cache **121**. The processor **110** may preload the image **823** first then preload the image **821** in response to the level_3 order of the image **821** being previous to the level_3 order of the image **823**.

[0079] The processor **110** may determine the number of the preloaded images of the directory **820** is '6', which is less than the count limit $A=15$. Therefore, the processor **110** may proceed to preload the other images. Since the offsets between the directory **820** and the directories **810** and **830** are within the range limit $B=1$, the processor **110** may preload the images within the directories **810** and **830**. The processor **110** may preload the images within the directory **830** first then preload the image within the directory **810** in response to the level_1 order of the directory **810** being previous to the level_1 order of the directory **830**. The processor **110** may preload the images **831** and **832** since the level_3 orders of those images satisfy the formula $1+K \leq \min(1+R, \text{the largest level}_3 \text{ order belonging to the } (i+I)\text{-th directory})$ as mentioned above. Then, the processor **110** may preload the images within the directory **810** since the number of the preloaded images is '10', which is less than the count value $A=15$. The processor **110** may preload the images **817** and **816** since the level_3 orders of those images satisfy the formula $\max(1, m-R) \leq m-K$ as mentioned above. The processor **110** may determine the number of the preloaded images is '14'. Although the number of the preloaded images '14' is less than the count value $A=15$, the difference between the count limit $A=15$ and '14' is less than the count value $C=2$. Accordingly, the processor **110** may stop preloading the images.

[0080] FIG. 9 illustrates a flowchart of a method for image viewing according to one embodiment of the present disclosure, wherein the method may be implemented by the electronic device **100** as shown in FIG. 1. In step **S901**, obtaining a plurality of images, wherein the plurality of images

comprise a first image, wherein the first image comprises a first value and a second value. In step S902, generating a directory of data by assigning the first image to a first directory of the directory of data and assigning the first image to a first subdirectory of the first directory according to the second value. In step S903, in response to the first image being selected, preloading a second image within a second directory of the directory of data to a cache memory according to the directory of data. In step S904, outputting a graphical user interface, wherein the graphical user interface displays the first image in response to the first image being selected.

[0081] In summary, the electronic device may create a directory of data according to values included in each of the image. When an image is being selected (or viewed) by the user, the electronic device may preload one or more images adjacent to the selected image in a specific sequence. For example, the electronic device may preload the image from the same video as the selected image, or the image within the same or adjacent directory (or subdirectory) as the selected image. Since the user is more likely to view the image later than the image previous to the selected image, the electronic device may preload the later image with a higher priority. If the selected image is close to the first or the last one in the directory, such that the fewer images need to be preloaded in the directory, the saved preloading quota can be used to preload the image within the adjacent directory of the selected image.

[0082] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. An electronic device for image viewing, comprising: a transceiver; a cache memory; and a processor, coupled to the transceiver and the cache memory, wherein the processor is configured to: obtain a plurality of images through the transceiver, wherein the plurality of images comprise a first image, wherein the first image comprises a first value and a second value; generate a directory of data by assigning the first image to a first directory of the directory of data according to the first value and assigning the first image to a first subdirectory of the first directory according to the second value; in response to the first image being selected, preload a second image within a second directory of the directory of data to the cache memory according to the directory of data; and output a graphical user interface through the transceiver, wherein the graphical user interface displays the first image in response to the first image being selected.
2. The electronic device according to claim 1, wherein the processor is further configured to: determine an offset between an order of the first directory in the directory of data and an order of the second directory in the directory of data; determine whether the offset is less than or equal to a first range limit; and in response to the offset being less than or equal to the first range limit, preload the second image within the second directory.
3. The electronic device according to claim 2, wherein the processor is further configured to: preload a third image within a third directory of the directory of data to the cache memory, wherein an order of the third directory in the directory of data is previous to an order of the second directory in the directory of data.
4. The electronic device according to claim 1, wherein the processor is further configured to: in response to the first image being selected, preload a fourth image within the first subdirectory to the cache memory according to the directory of data.
5. The electronic device according to claim 4, wherein the processor is further configured to: determine an offset between an order of the first image in the first subdirectory and an order of the fourth image in the first subdirectory; determine whether the offset is less than or equal to a second range limit; and in response to the offset being less than or equal to the second range limit, preload

the fourth image.

6. The electronic device according to claim 5, wherein the processor is further configured to: output a graphical user interface through the transceiver, wherein the graphical user interface displays a plurality of videos according to a count value.

7. The electronic device according to claim 6, wherein the processor is further configured to: determine the second range limit according to the count value.

8. The electronic device according to claim 5, wherein the processor is further configured to: in response to the offset being less than the second range limit but the fourth image being the first or the last one among images within the first subdirectory or in response to the first image being the first or the last one among the images within the first subdirectory, save a preloading quota; determine a number of preloaded images of the first directory; and in response to the number of the preloaded images being less than or equal to a count limit, preload the second images within the second directory by the preloading quota.

9. The electronic device according to claim 5, wherein the processor is further configured to: preload a fifth image within the first subdirectory to the cache memory after the fourth image being preloaded, wherein an order of the fifth image in the first subdirectory is previous to an order of the fourth image in the first subdirectory.

10. The electronic device according to claim 1, wherein the first image further comprises a third value, wherein the processor is further configured to: sort the first image in the first subdirectory according to the third value to generate the directory of data.

11. A method for image viewing, comprising: obtaining a plurality of images, wherein the plurality of images comprise a first image, wherein the first image comprises a first value and a second value; generating a directory of data by assigning the first image to a first directory of the directory of data and assigning the first image to a first subdirectory of the first directory according to the second value; in response to the first image being selected, preloading a second image within a second directory of the directory of data to a cache memory according to the directory of data; and outputting a graphical user interface, wherein the graphical user interface displays the first image in response to the first image being selected.

12. The method according to claim 11, wherein the step of preloading the second image within the second directory of the directory of data to the cache memory according to the directory of data comprising: determining an offset between an order of the first directory in the directory of data and an order of the second directory in the directory of data; determine whether the offset is less than or equal to a first range limit; and in response to the offset being less than or equal to the first range limit, preloading the second image within the second directory.

13. The method according to claim 12, further comprising: preloading a third image within a third directory of the directory of data to the cache memory, wherein an order of the third directory in the directory of data is previous to an order of the second directory in the directory of data.

14. The method according to claim 11, further comprising: in response to the first image being selected, preloading a fourth image within the first subdirectory to the cache memory according to the directory of data.

15. The method according to claim 14, wherein the step of preloading the fourth image within the first subdirectory to the cache memory according to the directory of data comprising: determining an offset between an order of the first image in the first subdirectory and an order of the fourth image in the first subdirectory; determining whether the offset is less than or equal to a second range limit; and in response to the offset being less than or equal to the second range limit, preloading the fourth image.

16. The method according to claim 15, further comprising: outputting a graphical user interface, wherein the graphical user interface displays a plurality of videos according to a count value.

17. The method according to claim 16, further comprising: determining the second range limit according to the count value.

18. The method according to claim 15, wherein the step of preloading the second image within the second directory of the directory of data to the cache memory according to the directory of data comprising: in response to the offset being less than the second range limit but the fourth image being the first or the last one within the first subdirectory or in response to the first image being the first or the last one among the images within the first subdirectory, saving a preloading quota; determining a number of preloaded images of the first directory; and in response to the number of the preloaded images being less than or equal to a count limit, preloading the second images within the second directory by the preloading quota.

19. The method according to claim 15, further comprising: preloading a fifth image within the first subdirectory to the cache memory after the fourth image being preloaded, wherein an order of the fifth image in the first subdirectory is previous to an order of the fourth image in the first subdirectory.

20. The method according to claim 11, wherein the first image further comprises a third value, wherein the method further comprises: sorting the first image in the first subdirectory according to the third value to generate the directory of data.
